Propeller Assembly – General Data

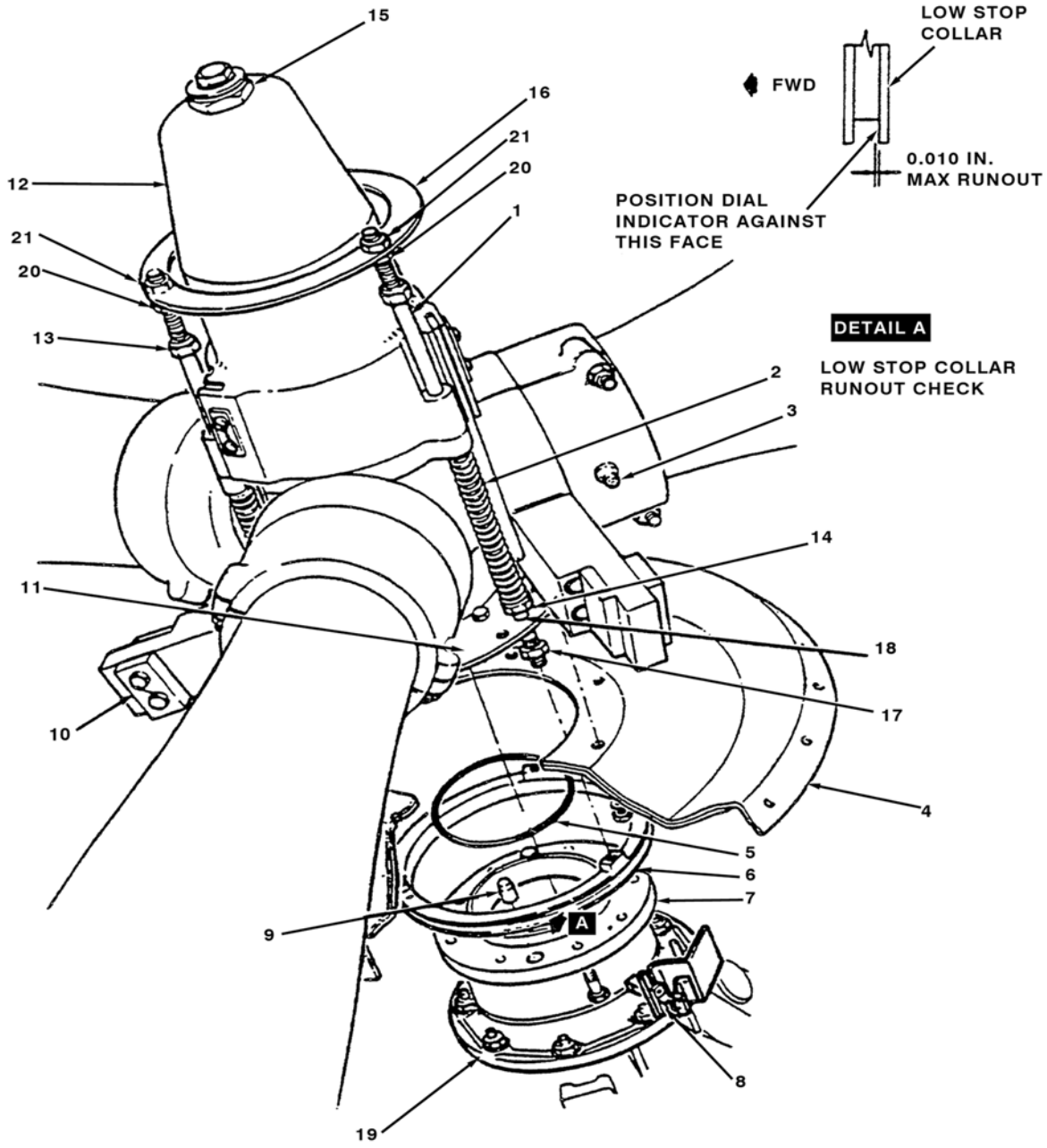
1. General

Refer to Figure 1 for details.

The propeller consists of a hollow spider hub which supports three propeller blades and houses the feather return springs. The propeller has a diameter of 8 feet 6 inches and a pitch range of 102 degrees, from –15 degrees reverse to +87 degrees feathered. When blade zero thrust stops are fitted, the zero stop setting is at +1 degree. The propeller is dowelled and bolted to the front face of the engine propeller shaft flange. Blade movement is controlled by a servo piston, mounted on the front of the propeller spider hub, which is connected by links to each blade root. During propeller operation, centrifugal counterweights attached to each blade, and the feather return springs in the spider hub, tend to move the blades into the high pitch or feather position, but this movement is opposed by oil pressure (controlled by the constant speed governor) acting on the piston. An increase in oil flow moves the blades towards the low pitch position (increased rpm), and relieving oil pressure allows the blades to move to the high pitch position, under the influence of the feather return springs and the blade counterweights. The piston is also connected to a low stop collar, mounted behind the propeller, by three spring loaded rods. Movement of the low stop collar in the beta range is relayed to a beta control valve (incorporated in the constant speed governor) by a carbon block and a propeller reversing lever, to control blade angles from the normal forward low pitch stop to full reverse.

Propeller blade zero thrust stop assemblies may be fitted by the propeller manufacturer for operation on floats. (Refer to 61-26-00, Blade Zero Thrust Stop System (S.O.O. 6203) – General Data).

On special order, a propeller de-icing system may be installed. (Refer to 61-10-11, Propeller De-icing without Pitch Latches (S.O.O. 6202) – General Data or 61-10-12, Propeller De-icing with Pitch Latches (S.O.O. 6237) – General Data).



1. LOW PITCH STOP ROD
2. REVERSE RETURN SPRING
3. LUBRICATION FITTING
4. SPINNER BULKHEAD
5. O-RING
6. LOW STOP COLLAR
7. ENGINE SHAFT FLANGE
8. MICROSWITCH SLIDE BLOCK
9. DOWEL PIN
10. COUNTERWEIGHT
11. SPINNER SUPPORT RING
12. PISTON UNIT
13. PITCH SETTING NUT
14. STOP ROD WRENCH FLATS
15. FLEXLOCK NUT
16. ROD END RING
17. LOCKNUT
18. ROD GUIDE
19. ENGINE REDUCTION GEAR CASE
20. ROD END RING REAR LOCKNUTS
21. ROD END RING FORWARD LOCKNUTS

Propeller Installation
Figure 1

2. Removal/Installation

Refer to Figure 1 for details.

Equipment Required:

- Sling SD5534
- Wrench SD12540
- Retraction Clamp SD12559
- Wrench SD12549
- Container to receive oil spillage

A. Remove Propeller

1) Remove spinner, nut and bolt securing teleflex control clevis rod end to propeller reversing lever, and pin from pivot at beta control valve clevis slot, retain spacer and bushing. Disengage and remove reversing lever complete with carbon block from low stop collar.

2) Attach sling (SD5534) to propeller (refer to 07-00-00, Lifting and Shoring – General Data) and support weight.

3) Position container to receive oil spillage.

4) If retraction clamp (SD12559) is to be used during removal, proceed as follows:

CAUTION **DO NOT DISTURB PITCH SETTING NUTS (13) OR THE REAR ROD END RING LOCKNUTS, FOR THE ADJUSTMENT OF THE PITCH SETTING NUTS. REFER TO 71-00-00, POWER PLANT (MOD 6/2012) – GENERAL DATA ADJUSTMENT/TEST.**

a) Remove the three forward locknuts securing rod end ring (16). Do not allow stop rods to rotate.

b) Remove rod end ring (16).

c) Using retraction clamp, screw clamp bolts onto ends of low pitch stop rods. Do not allow stop rods to rotate. Insert clamp socket over hexagon on piston unit ensuring that clamp bolts are evenly adjusted. Rotate center tee-bolt to pull low pitch stop rods and low stop collar forward to allow access to propeller retaining bolts.

d) Using wrench (SD12540) remove propeller retaining bolts and remove propeller. Retain the beta microswitch carbon block and slide which becomes detached from switch mounting as propeller is moved forward.

e) Remove propeller sling and retraction clamp.

f) Screw two of the three rear rod end ring locknuts further onto low stop rods; do not disturb third rear locknut. Install rod end ring (16) and secure with one locknut at the undisturbed locknut position. Turn the two rear rod end ring locknuts (previously screwed further onto rods) until they just contact the rear face of the rod end ring; install the two forward locknuts at these positions. Tighten all three forward locknuts. Do not allow stop rods to rotate.

5) If retraction clamp is not to be used during a propeller removal, proceed as follows:

a) Remove flexlock nut (15) using wrench (SD12549).

b) Rotate all blades simultaneously slowly to full reverse. Ensure locknuts do not foul guide rod lugs. Secure blades in this position.

c) Install sling SD5534 to support propeller.

d) Using wrench (SD12540) remove propeller retaining bolts and remove propeller. Retain the beta microswitch carbon block and slide which becomes detached from switch mounting as propeller is moved forward.

e) Remove propeller sling and install flexlock nut (15), ensuring O-ring seal is in position. Using wrench (SD12549) torque flexlock nut to 120 pound–feet.

B. Install Propeller

1) Attach sling (SD5534) to propeller and raise into position in front of engine shaft. (Refer to 07-00-00, Lifting and Shoring – General Data).

2) Ensure propeller and engine flanges are clean and free from scoring or burring which could prevent mating.

3) If retraction clamp (SD12559) is to be used, position low stop collar as detailed in Para 2.A.4).a), and as applicable Para 2.A.4).c) or Para 2.A.4).d).

4) If retraction clamp is not to be used, position low stop collar as detailed in Para 2.A.5).a) and Para 2.A.5).b).

5) Install propeller according to Hartzell Owner's Manual 139 (61–00–39) (latest revision) using wrench (SD12540), or (SD12513) if retraction clamp is not being used.

3. Inspection/Check

A. Check Propeller Low Stop Collar Runout

1) Ensure propeller is feathered.

2) Suitably mount a dial indicator on engine reduction gear case.

3) Rotate propeller slowly and check runout of working face of low stop collar.

Runout must not exceed 0.010 inch.

Note If runout exceeds 0.010 inch, the collar may be adjusted by slackening the low stop locknuts adjacent to collar, rotating the low stop rods to adjust insertion of rod in collar and retightening locknuts. Torque load locknuts to 12 pound–feet.

CAUTION IF THE LOW STOP RODS ARE ADJUSTED, THE PITCH SETTING NUTS MUST BE READJUSTED (REFER TO 71-00-00, POWER PLANT (MOD 6/2012) – GENERAL DATA, ADJUSTMENT/TEST).